

Speaking, writing, and memory span in children: Output modality affects cognitive performance

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Low-level processes of children's written language production are cognitively more costly than those involved in speaking. This has been shown by French authors who compared oral and written memory span performance. The observed difficulties of children's, but not of adults' low-level processes in writing may stem from graphomotoric as well as from orthographic inadequacies. We report on five experiments designed to replicate and expand the original results. First, the French results were successfully replicated for German third-graders, and for university students. Then, the developmental changes of the cognitive costs of writing were examined during primary school, comparing the performance of second- and fourth-graders. Next, we show that unpractised writing modes, which were experimentally induced, also lead to a decrease of memory performance in adults, which supports the assumption that a lack of graphomotoric automation is responsible for the observed effects in children. However, unpractised handwriting yields clearer results than unpractised typing. Lastly, we try to separate the influences of graphomotoric as opposed to orthographic difficulties by having the words composed through pointing on a "spelling board". This attempt, however, has not been successful, probably because the pointing to letters introduced other low-level costs. In sum, throughout the four years of primary school, German children show worse memory span performance in writing compared to oral recall, with an overall increase in both modalities. Thus, writing had not fully caught up with speaking regarding the implied cognitive costs by the end of primary school. Therefore, conclusions relate to the question of how to assess properly any kind of knowledge and abilities through language production.

Los procesos de bajo nivel en la producción de lenguaje escrito en niños son más costosos a nivel cognitivo que los que están implicados en el habla. Esto ha sido demostrado por autores franceses quienes compararon la ejecución en la capacidad de retención en la memoria oral y escrita. Las dificultades observadas en niños, pero no en adultos, en procesos de bajo nivel en la escritura pueden derivarse de la inadecuación grafomotriz u ortográfica. Informamos sobre cinco experimentos diseñados para replicar y ampliar los resultados originales. Primero, los resultados franceses fueron replicados con éxito para alemanes en tercer curso y para estudiantes universitarios. Después, los cambios evolutivos en costes cognitivos en la escritura fueron estudiados durante la educación primaria, comparando la ejecución de los alumnos del segundo y cuarto curso. Posteriormente, demostramos que las formas de escribir no practicadas, que se han inducido experimentalmente, conducen a la disminución de la ejecución en la memoria también en adultos, lo cual apoya la hipótesis de que la falta de la automatización grafomotriz es la responsable de los efectos encontrados en niños. Sin embargo, la escritura a mano muestra resultados más claros que la escritura a máquina. Por último, hemos intentado separar la influencia de la grafomotricidad frente a las dificultades ortográficas componiendo las palabras señalando en una "pizarra de ortografía". Este intento, sin embargo, no ha tenido éxito, probablemente porque señalar letras ha introducido otros costes de bajo nivel. Resumiendo, a lo largo de los cuatro años de la educación primaria, los niños alemanes muestran peor ejecución en retención en memoria en la escritura, en comparación con el recuerdo oral, con un incremento general en ambas modalidades. Por tanto, al final de la educación primaria, la escritura no ha igualado todavía el habla en cuanto a los costes cognitivos. Por consiguiente, las conclusiones están relacionadas con la cuestión de cómo evaluar cualquier tipo de conocimiento o habilidades mediante la producción de lenguaje.

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Les processus de bas niveau de la production du langage écrit chez les enfants sont cognitivement plus coûteux que ceux impliqués dans le langage parlé. Ceci a été démontré par des auteurs français qui ont comparé la performance de l'étendue de la mémoire orale et écrite. L'observation de difficultés dans les processus de bas niveau dans l'écriture chez les enfants, et non chez les adultes, pourrait être due aux inadéquations graphomotoriques et orthographiques. Nous rapportons cinq expérimentations dont le but était de répliquer et d'élargir les résultats originaux. Premièrement, les résultats français ont été répliqués avec succès pour les élèves allemands de troisième année et pour les étudiants universitaires. Ensuite, les changements développementaux des coûts cognitifs de l'écriture ont été examinés à l'école primaire en comparant la performance des élèves de deuxième et de quatrième années. Par la suite, nous avons montré que les modes d'écriture non pratiqués qui étaient expérimentalement induits indiquent une diminution de la performance de la mémoire, chez les adultes aussi, ce qui supporte la supposition qu'un manque d'automatisme graphomoteur est responsable des effets observés chez les enfants. Cependant, une écriture à la main sans entraînement mène à des résultats plus clairs que la dactylographie sans entraînement. Enfin, nous avons tenté de séparer les influences des difficultés graphomotoriques versus difficultés orthographiques à travers la composition des mots en les indiquant sur un tableau d'orthographe. Cependant, cette tentative n'a pas réussi, probablement à cause du fait d'indiquer les lettres présente d'autres coûts dont le niveau est bas. En résumé, à travers les quatre années de l'école primaire, les enfants allemands montrent une pire performance de l'étendue de mémoire dans l'écriture, en comparaison au rappel oral, avec une augmentation globale dans les deux modalités. Ainsi, toujours à la fin de l'école primaire, l'écriture n'a pas rattrapé l'expression orale en ce qui concerne les coûts cognitifs implicites. Par conséquent, les conclusions se rapportent à la question de comment évaluer correctement n'importe quel type de connaissances et d'habiletés à travers la production du langage.

Keywords: Language production; Writing development; Working memory; Memory span.

Bourdin and Fayol (1994) started to report on research that examined low-level activities in writing, asking whether oral or written language production is more difficult for children and adults. The distinction between high-level and low-level processes is characteristic of hierarchical theoretical accounts of language production referring to a general-resource model of the cognitive system, where information processing proceeds within limited working memory capacity (see Berninger, Fuller, & Whitaker, 1996; McCutchen, 1995; Torrance & Galbraith, 2006, for general accounts of cognitive processes in writing and writing development).

While high-level or planning processes involve the generation, selection, and ordering of ideas, low-level activities relate to lexical access and sentence generation (formulation), and to graphic transcription and execution in writing, or phonological encoding and articulation in speaking, respectively. Bourdin and Fayol assume that both high-level and low-level components of writing processes put a load on working memory, and that with increasing load by one component, performance based on other components suffers due to decreases in the remaining resources (Just & Carpenter, 1992). Further, the authors assume that the low-level processes of writing are not yet fully automated in children, particularly with respect to spelling and the motor activity of

handwriting. Thus, performance of higher-level activities should be impaired in children's writing, but not in adults whose handwriting skills are well practised.

While this approach adopts a paradigmatic perspective of general psychology, explaining memory performance from different language production modes, there have also been differential approaches to the connection between working memory and writing. For example, Swanson and Berninger's (1996) work is aimed at predicting writing and reading measures from individual differences between working memory components. In the following, however, we will further pursue and elaborate on the general approach of Bourdin and Fayol, who intended to specifically address low-level processes by experimental means. For that, they used a task which did not call for planning, idea generation, or sentence formulation, excluding high-level processes and isolating low-level processes as much as possible.

Participants were acoustically provided with word lists of increasing length. After presentation of each list, they were to recall all words in the original order. This task was given in the oral and written recall modes. Since all but the lowest-level processes (orthographical vs. phonological encoding and graphomotoric vs. articulatory execution) are identical in speaking and writing, differences between the two modalities are attributable to

different amounts of load on the available cognitive resources. In their main experiment, university students, second-graders, and fourth-graders were presented with word lists of different lengths. There were two versions of each list, one for oral and one for written recall. Participants constantly changed recall modes with increasing list lengths.

Results include no mode effect for university students (serial recall remained consistent across written and oral recall); a significant mode effect for both groups of pupils (better recall in the spoken than in the written mode); and a significant effect of the children's age (fourth-graders generally do better than second-graders). These results indicate that in second-graders and in fourth-graders writing is not yet fully automated and involves higher cognitive load than speaking, thus reducing working memory capacity available for the memory task.

Since a reformulation of the given interpretation in terms of more recent explanatory accounts of limited processing resources in working memory (Cowan, 2005; Oberauer & Kliegl, 2006) would not change the essential meaning of the results, we retain the notion of capacity in the tradition of Just and Carpenter (1992).

Subsequently, Bourdin and Fayol (1994, 2000, 2002) added series of experiments to support their interpretation against several possible objections. From these experiments, we learn that it is not the faster execution speed of speech that facilitates the recall task as such, because the result pattern remains stable in fixed-recall conditions where oral recall was slowed to the median writing rate. When the transcription of the recalled items was done by the experimenter (dictation), children performed similarly as in the oral mode, whereas performance of adults decreased when forced to use the unpractised script of cursive capital letters, suggesting that a lack of automation in handwriting could be responsible for the overall result pattern. While all of the results discussed so far should apply to members of all linguistic communities (at least with alphabetic script), because their interpretations relate to general cognitive processes, the authors continue their series of experiments by considering aspects specific to French, like the more or less regular spelling of words and their familiarity. Here, however, we will concentrate on general cognitive implications rather than the peculiarities of orthographies (Perfetti, Rieben, & Fayol, 1997).

There are several reasons why the above-reported research still deserves attention and further scrutiny. First, the complexity of language

production processes is often beyond systematic experimental control. Research on language comprehension processes is, historically and quantitatively, much more advanced, because the materials are easier to control (Bock, 1996) and measures of understanding are easier to obtain and analyse in the form of closed-class variables. On the contrary, Bourdin and Fayol found appropriate experimental variations to decompose language production such that the interesting aspects remain under full experimental control. With a serial recall task, the authors nicely eliminated any variation due to planning processes, syntactic processes, and processes of lexical selection (see Brown, McDonald, Brown, & Carr, 1988, for a similar strategy). The remaining processes and their explanation, however, still appear complicated enough to be further pursued.

Second, after an early period of investigation in which speaking and writing have been compared with respect to their products or their temporal economy (e.g., Gould, 1978), both modes of language production have been mainly considered separately. It would expand our knowledge, however, to compare directly the functioning of both verbal modes on all processing levels involved. Thus, it is far from clear whether speaking and writing offer equivalent possibilities for students to express their cognitive achievements (Grabowski, 2007). Here, Bourdin and Fayol offer a theoretically uniform approach to the comparison of oral and written language production on low-level processes.

Reading Bourdin and Fayol's results from an educational perspective, they point to a broadly neglected problem: the role of language production for diagnostic processes in primary school. If, as their results show, speaking and writing differ with respect to the loads imposed on working memory resources and, therefore, lead to significantly different performances on simple tasks such as serial recall of word lists, it is doubtful whether both verbal modes would provide equivalent indications of more complex cognitive processes in long-term memory, where many processes are involved in which speaking and writing could differ in relevant ways (Grabowski, 2007).

Altogether, the results appear relevant enough to make it worth finding out whether they also emerge in populations different from French speakers. If the authors' interpretation is correct, their findings should allow for successful replication among the speakers of another language, because the observed differences between the oral and written modes relate to general cognitive principles. On the other hand, French is a language with a rather irregular orthography, in

part due to its many silent graphemes. Consistency between the phonological (sounds) and the orthographical (letters) segments is low. The predictability from the phonological to the orthographical composition of words is 50.1% for German, but only 20.9% for French (R. Graf, personal communication). Predictability is the proportion of words that have, given their phonological forms, only one possibility of graphic realization. (Interestingly, consistency from the orthographical to the phonological form is close to 90% in German and French; Jacobs & Graf, 2005.) Therefore, it could be that a more predictable orthography like German (Wimmer, 1995) would pose fewer difficulties for the cognitive systems of young writers.

EXPERIMENT A: REPLICATION OF THE MAIN EXPERIMENT

Experiment A examines the expectation of Bourdin and Fayol (1994, p. 597), according to which in children, but not in adults, writing puts a greater load on working memory than speaking. Like the French original, we used a word-span test (see Daneman & Carpenter, 1980). However, Bourdin and Fayol (1994, p. 599) did not assess individual memory spans, but analysed the proportions of correctly recalled words for lists with proportions close to .50. We will not adopt their analysis for three reasons. First, it is statistically problematic: It compares proportions of different things (about 50 per cent of lists comprising seven words, five words, or four words), and if a group were empirically better or worse, another list would serve as the basis of comparison. It doesn't become visible in Bourdin and Fayol's analyses that the absolute performance levels of the groups substantially differ. Second, this analysis doesn't connect to memory span measures, so that meaningful comparisons across experiments or age groups, and also with respect to other authors, remain impossible. Third, it takes a standardized measure such as memory span to compare results on serial recall tasks across materials and procedures (van den Noort, Bosch, Haverkort, & Hugdahl, 2008). Therefore, we will assess simple word spans according to Daneman and Carpenter (1980): the number of words that, after hearing them once, can be recalled in the correct order.

Method

Participants

Two samples of participants took the test: $N = 24$ third-graders of a German primary school (11 girls;

mean age: 9 years, 6 months, ranging from 9 years, 0 months to 10 years, 3 months) and $N = 40$ teacher students from Heidelberg University of Education (seven male; mean age 23 years). Concerning age, German third-graders correspond to Bourdin and Fayol's fourth-graders; their mean age was 9 years, 7 months.

Materials

Two hundred and fifty-two frequent bisyllabic German words, five to eight letters in length, were selected from the CELEX database (Baayen, Piepenbrock, & van Rijn, 1993) and divided into two lists containing word sets from three to nine words (= set size). There were three sets of each size in both lists. Sets three to seven words long were used for the children sample and sets four to nine words long for the university students sample. Words were randomly assigned to lists and sets.

Procedure

Participants were tested individually in a separate room of their school or university. After a general instruction, a practice trial with three sets of two words each made participants familiar with the recall task. All sets were presented orally by the experimenter, at a rate of one word per second. Participants were instructed to start recall right after they heard the complete set. In the oral mode, the experimenter compared the participants' recall with a control list and indicated correct or false responses. In the written mode, a booklet had been prepared containing subsequent pages for each set of words.

Sets from one list were taken for oral recall; sets from the other list for written recall. Half of the participants in each sample began with the oral condition; the other half with the written condition. Each condition started with the shortest set size (three words for children, four words for students) and continued until none of the three sets of a given size was successfully recalled. Assignment of lists to conditions and to their order in the experiment was systematically counterbalanced.

This procedure differs in two respects from the French original. First, Bourdin and Fayol (1994) switched between oral and written recall on each set length: Their participants were to prepare themselves for another recall mode after every set. Particularly for the children's sample, it could have additional costs to remember which kind of recall is expected next. Therefore, we ran a whole mode condition, from the shortest to the longest set size, and then repeated the entire procedure in the other

mode. Second, Bourdin and Fayol tested recall performance only by one set of words of a given length within each condition. This appears too susceptible to attentional distractions. We used three sets of each size, which also facilitates the calculation of span measures according to existing standards.

Analysis

For both conditions, the word-span score assigned to an individual's performance is the highest size, at which all three sets have been correctly recalled. "Correct" means to recall the original morphological word forms in the original order. If one of the three sets of the next size has also been correctly recalled, 0.5 is added to the score. If two sets of the next size have been correctly recalled, 1.0 is added. However, correctly recalled sets within a set size are only considered if all sets of the previous size have been successfully recalled.

Results and discussion

There were no effects due to the assignment of lists to conditions or to order of conditions within a session; thus, control conditions were grouped together. Word-span means of the four experimental conditions were as follows (standard errors given in parentheses): children/oral 3.56 (.13); children/written 3.13 (.11); adults/oral 5.06 (.10); adults/written 5.11 (.12). Results are illustrated in Figure 1. An ANOVA with "oral vs. written span" as within-subjects factor and "children vs. adults" as between-subjects factor revealed a significant main effect for age, $F(1, 62) = 152.5$, $MSE = .598$, $p < .0001$, and a significant interaction between recall mode and age, $F(1, 62) = 7.88$, $MSE = .226$, $p < .01$. When one is comparing oral and written spans in pairs within the two age groups, oral word span is higher than written word span in

children, $t(23) = 3.99$, $p < .001$, but not in adults, $t(39) = -0.43$, $p > .5$.

The obtained findings match those obtained from French participants in Bourdin and Fayol's (1994) study in three relevant aspects. (1) German primary school children obviously experience loads on working memory comparable to French children of that age, due to their not yet fully automated writing skills. (2) For adult university students with largely automated writing skills, no difference between spoken and written output modes occurred, indicating that both verbal modalities involve low-level processes with low cognitive costs. (3) University students perform better than primary school children in terms of absolute span sizes.

EXPERIMENT B: REPLICATION OF THE DEVELOPMENTAL EFFECT ACROSS PRIMARY SCHOOL

In addition to the key finding of increased cognitive load of low-level writing processes in children, Bourdin and Fayol (1994, Experiment 1) also found that fourth-graders performed better than second-graders on both verbal modes of the recall task. Surprisingly, no age $\times$ mode interaction appeared, indicating that primary school children develop, from second to fourth class, some general cognitive ability, but that automation of low-level writing skills doesn't substantially catch up to come closer to oral performance. Would this picture also emerge for German primary schoolers?

Method

Participants

Students of two classes from different schools, different from Experiment A participants, performed word-span tests in the oral and written mode. $N = 16$ children were in second grade (six

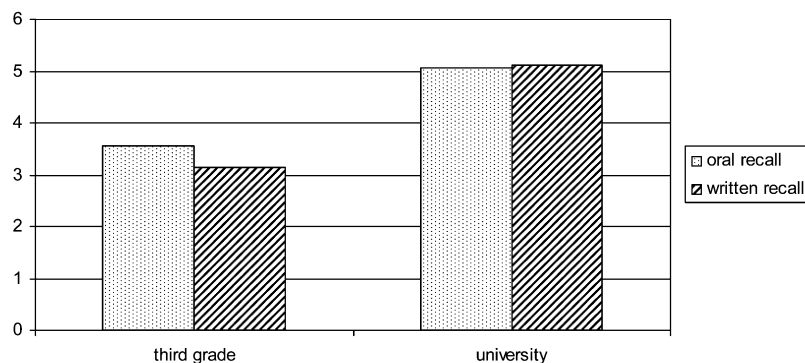


Figure 1. Experiment A: Mean oral and written word span by age.

boys; mean age: 7 years, 10 months, ranging from 6 years, 11 months to 8 years, 5 months). $N = 20$ children were in fourth grade (12 girls; mean age: 10 years, 0 months, ranging from 9 years, 5 months to 10 years, 10 months).

Materials

One hundred and fifty frequent bisyllabic German words, five or six letters in length, were selected from the CELEX database (Baayen et al., 1993) and divided into two lists containing word sets from three to seven words (= set size), three sets per size in each list. Words were randomly assigned to the lists and sets, except for cases in which obvious semantic relations between words within the same set were removed.

Procedure and analysis

Students were tested individually in a separate room of their school. Procedure and data analysis correspond to Experiment A, including counter-balance of mode order and list assignments to modes.

Results and discussion

There were no effects due to the assignment of lists to conditions or to the order of conditions within a session; thus, control conditions were grouped together. Word-span means of the four experimental conditions were as follows (standard errors given in parentheses; see Figure 2): second-graders/oral 3.81 (.15); second-graders/written 2.84 (.13); fourth-graders/oral 4.30 (.11); fourth-graders/written 3.63 (.10).

An ANOVA with "oral vs. written span" as within-subjects factor and "second vs. fourth grade" as between-subjects factor revealed a significant main effect for age, $F(1, 34) = 19.1$, $MSE = .375$,

$p < .001$; fourth-graders performed better than second-graders. Another significant main effect of recall mode emerged, $F(1, 34) = 87.1$, $MSE = .138$, $p < .001$; oral performance was consistently better than written performance.

As in the French study, there is no significant mode $\times$ age interaction, $F(1, 34) = 2.78$, $MSE = .138$, $p > .10$, indicating an increase in both oral and written recall from second to fourth grade, but no substantial decrease from second to fourth grade regarding the difference between oral and written performance. Thus, until the end of primary school, low-level processes of writing put load on working memory resources, so that it may still make a difference whether performance is assessed through oral or written language production.

EXPERIMENTS C1 AND C2: RAISING THE MENTAL COSTS OF WRITING IN ADULTS

If insufficient automation of motor processes of handwriting causes increased cognitive load, which in turn impairs memory span performance, then it should be possible to produce similar effects artificially in adults despite their automated writing skills: If the writing situation is altered such that well-practised routines are interrupted, memory span should decrease. In the Bourdin and Fayol series of experiments, this has been demonstrated for handwriting. Below, we report on a replication of this study (Experiment C1) and an elaboration for typing (Experiment C2).

For handwriting, we adopted the methodical strategy of Bourdin and Fayol to prohibit participants from using their familiar handwriting style, or print types, forcing them into extremely neat cursive writing. However, motor execution and its possible cognitive load occur not only with handwriting, but also with typing. While fluent handwriting requires the automated drawing of

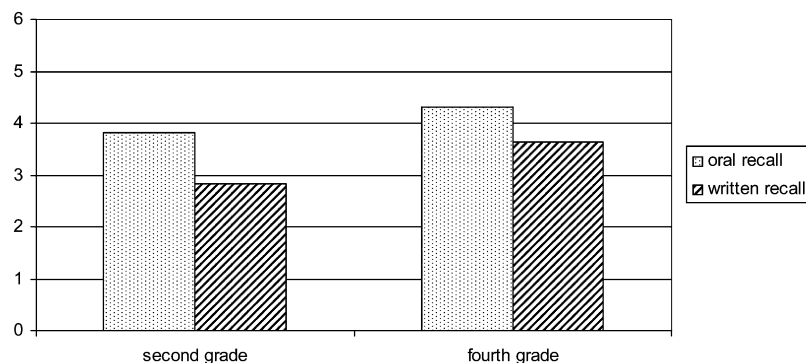


Figure 2. Experiment B: Mean oral and written word span by age.

stroke sequences and the connections between them (Thomassen & Van Galen, 1992), fluent typing (whether or not a touch-typing method is being mastered) requires the automation of well-directed finger movements to different locations on the keyboard (Rieger, 2004). It is doubtful, however, whether there is enough variation of typing skills among university students to justify a subdivision of practised and unpractised keyboard writers (Grabowski, 2008). Therefore, we considered all participants as experienced typers, and experimentally induced a condition of unpractised typing through the arbitrary exchange of keys on the keyboard. If decrease in memory span performance is attributable to increased cognitive load through any unpractised motor components of writing, it should also emerge in adults with unfamiliar handwriting and unfamiliar typing.

Method of experiment C1

Participants

$N = 30$ psychology students of the Catholic University of Eichstätt (six male; mean age: 22 years, 4 months) were assessed for their memory spans, after they were randomly assigned to either the “practised” ($n = 15$) or the “unpractised” ($n = 15$) handwriting group. In contrast to the above-mentioned experiments, variation of recall modes followed a between-subjects design.

Materials

One hundred and fifteen frequent, bisyllabic German words, five to seven letters long, were selected from CELEX database (Baayen et al., 1993) and randomly grouped to sets of five to nine words, three sets in each instance. Two five-word sets were prepared as practice trials. Both experimental groups received the same word lists. Each set was electronically recorded with a female voice reading the respective list at a pace of one word per second, with a signal tone at the beginning and at the end of each set. The sound files were arranged in a PowerPoint presentation according to the required sequence.

For the “unpractised writing” condition, an alphabetically ordered pattern of capital cursive letters, which also illustrated the appropriate direction of sweep, was shown to the participants before they started.

For each participant, a booklet was prepared containing one page for each set, with empty lines for the number of words in the respective set.

Procedure and analysis

Participants were tested individually. Participants in the “unpractised writing” condition were first asked to copy once the stipulated letters of the pattern. Instructions were given on a computer screen. Then, participants put headphones on and proceeded through the experiment at their own pace, triggering the next set of words by a mouse click. According to the experimental group, they recalled the word lists either in their normal handwriting script or in capital cursive letters. Notwithstanding success or failure on previous sets, all participants were to proceed to the very end (nine-words lists).

Results of experiment C1

A t -test revealed a statistically significant difference between the achieved memory spans in the two conditions of practised ($M = 5.87$, $SE = .12$) and unpractised ($M = 5.30$, $SE = .15$) handwriting, $t(28) = 2.88$, $p < .01$. Expectedly, an unpractised mode of handwriting causes a decrease of the cognitive resources available for the retention of verbal materials.

Method of experiment C2

Participants

$N = 30$ psychology students of the Catholic University of Eichstätt (20 female; mean age: 26 years) were assessed for their memory spans, after they were randomly assigned to either the “familiar” ($n = 15$) or the “unfamiliar” ($n = 15$) keyboard condition (between-subjects design).

Materials

Word lists were prepared and recorded as in Experiment C1, but with different words. Text input software was programmed that had an input window at the bottom and nine lines in the upper half of the computer screen. Each remembered word was to be entered in the input window and acknowledged with the enter key, after which it was copied to the uppermost empty line, and the input window was emptied. Before a participant listened to a new set, all words of the previous set were removed from the screen. Data files were automatically generated.

For the “unfamiliar” keyboard condition, the software exchanged, in pairs, ten keys before their letter values appeared on the screen: “o” and “e”; “h” and “r”; “b” and “t”; “n” and “a”; “l” and

“s”. Obviously, the most frequently used keys were among this selection. The keyboard surface was changed such that the exchanged keys were marked with their new values. Thus, all participants of this group were forced to display the typing behaviour of an unpractised novice.

Procedure and analysis

Except for the writing mode and the presentation mode (the recorded word lists were replayed by control of the experimenter rather than by control of the participants), procedure and analysis corresponded to Experiment C1.

Results of Experiment C2

A *t*-test revealed a trend in the expected direction: The mean memory span was higher in the “familiar” ($M = 6.6$, $SE = .37$) than in the “unfamiliar” keyboard condition ($M = 5.97$, $SE = .39$). Statistical significance failed, $t(28) = 1.19$, $p > .10$, due to comparatively large standard errors in both conditions, although the absolute difference of the spans is higher than in experiment C1.

Did the varying mastery of a touch-typing method influence the difficulty of the task? Those participants who said in a questionnaire that they use a touch-typing method (“10 fingers”) didn’t show better overall performance than participants without such a method, $M = 6.21$, $SE = .38$ vs. $M = 6.34$, $SE = .38$; $t(28) = 0.24$, *ns*. (Note that the members of the two groups were equally frequent within the two keyboard conditions.)

Results of experiments C1 and C2 combined

Since experiments C1 and C2 have identical designs and sufficiently corresponding procedures and materials, their combined data can be analysed in an ANOVA with “writing mode” (handwriting vs. typing) and “practice” (practised vs. unpractised writing mode) as independent factors. This analysis yields a main effect for writing mode, with typing allowing for better memory span performance than handwriting, $F(1, 55) = 5.74$, $MSE = 1.23$, $p < .05$. Moreover, there is a main effect for practice in favour of the respective better practised writing modes, $F(1, 55) = 4.20$, $MSE = 1.23$, $p < .05$, and no interaction, $F(1, 55) = 0.01$.

Discussion of experiments C1 and C2

Writing, whether by hand or keyboard, displays the characteristics typical of any complex behaviour

with relevant motor components: It allows the cognitive powers of memory, problem-solving, or decision-making to come into effect only if the basic motor components of the respective task are sufficiently automated (Moors & De Houwer, 2006). Even for the simple writing tasks presented in our experiments (which disregard higher-level processes such as planning, syntactic formulation, and choice of words for concepts), cognitive performance can be crucially impaired by an unpractised writing mode. This points to the importance of a successful acquisition of basic (hand)writing skills during school for any further education, but also for reconsideration of the role of keyboard writing in higher education (which is often required, and employed, without much previous motor practice; see Grabowski, 2008). In the two experiments, however, a statistically significant effect of unpractised writing modes has only been given proof for handwriting and for a *post-hoc* recombination of the two experiments. Apparently, with young adults there are more individual differences in the automation of keyboard operation than in handwriting automation, which leads to high standard variations and, in turn, prevents statistical significance despite a big difference between the respective memory span means.

In sum, we assume that any language production task that possibly involves poorly automated components (not only writing modes, but also the phonology of a foreign language, for example) must be examined carefully for “hidden” effects of low-level processes.

Since we have shown that, in adults, memory span can decrease when the motor components of handwriting (and probably also of typing) are made more difficult, we may reason that it is unpractised motor behaviour in children that contributes to the observed differences between oral and written memory spans. However, in order to write down the words retained in memory, it is also necessary to involve spelling skills. In order to write down a word, we must apply either some orthographic rule or a phonological analogy, both involving spelling processes. Therefore, the last experiment is aimed at separating the cognitive costs of graphomotoric behaviour from those of spelling.

Note, however, that we must acknowledge that low-level processes of writing comprise motor execution and spelling as such, but not necessarily correct spelling. Whenever a word is written down, it must be spelled somehow. Thus, mastery of orthography is a relief, or simplification, of the writing process because the writer need not run an explicit decision process in order to find a letter

representation for a sound sequence. (Think of the difficulties of expressing dialects or accents in writing.) In primary school children, handwriting abilities and (correct) spelling abilities do not necessarily covary. Graham, Berninger, Abbott, Abbott, & Whitaker (1997), for example, report a rather low correlation of .22 between handwriting (measured by a copy task) and a spelling test.

EXPERIMENT D: DIFFERENTIATING SPELLING FROM GRAPHIC ACTIVITY

In order to release children from graphic activity, Bourdin and Fayol (1994, Experiment 3a) used a dictation mode, where the experimenter transcribed the remembered words that were dictated to them. According to the authors, the simplification of the task didn't go so far that children would also be released from any spelling-related processes. Rather, the "children checked the accuracy of the transcription" (p. 605). However, there was no systematic control of these checks (B. Bourdin, personal communication), for example by instructing the experimenter to misspell words at a given rate. It turned out that the children's memory spans in the dictation mode were as high as in the oral mode, and compared to both conditions, writing again led to a decrease of memory span performance. We cannot know from the experimental procedure, however, whether the children in Bourdin and Fayol's dictation mode did anything different than in the oral mode with respect to the cognitive costs of the involved processes. Other research using dictation tasks in order to try to disentangle the subcomponents of low-level process in writing (e.g., Scardamalia, Bereiter, & Goleman, 1982) was interested in the effects of different production modes on the quality and/or quantity of written composition (see Graham et al., 1997, p. 171, for an overview), but not directly on working memory.

We tried to introduce a systematic differentiation between spelling and graphomotoric costs by the use of a "spelling board". On a piece of cardboard, about 50 × 50 cm, we stuck (in an array of five lines with six letters each) the 26 letters of the German alphabet in block capitals plus three capital umlaut letters and the special sign for a double "s" after long vowels (which only exists as a lower-case letter). After they heard a word list of the respective length, children were to silently "write" or "spell", respectively, the remembered words by pointing to the successive letters of which the written words are composed (see Figure 3). Pointing is a graphic activity much

simpler than handwriting, and the requested simple movements are the same for all letters, whereas the spelling component of the task is fully maintained. The more the graphic activity of handwriting is responsible for the lower memory spans in writing (as reported from Experiments A and B), the more should the results of the pointing mode equal the results of the oral mode. Moreover, composing words by pointing to letters will presumably not take longer than handwriting; rather, pointing to a letter is accomplished faster than drawing a letter. However, pupils are not very practised in locating letters in a 5 × 6 array, although alphabetically ordered. Therefore, the "search" for the location of respective letters on the spelling board could also have the side-effect of a task that uses up central working memory resources which would lead to a decrease of memory performance. In this case, the spelling board would have effects on children's memory performance similar to those observed for adults in Experiments C1 and C2, where unpractised writing modes were introduced.

Method

Participants

N = 17 fourth-graders (12 boys; mean age: 9 years, five months, ranging from 8 years, 10 months to 9 years, 11 months) took memory-span measures in three conditions each: "speaking", "writing", and "pointing" (= within-subjects design). The order of the three conditions was systematically varied between participants. Two other girls were excluded from the analysis,



Figure 3. A pupil recalling a word on the "spelling board".

because they were not able to recite the alphabet correctly within 30 s, which was set as a criterion. If a child were not even able to recite the alphabet fluently, it would be clear from the outset that he or she would not be able to profit from any array of letters based on their alphabetic order. For the remaining 17 participants, the time needed for correctly reciting the alphabet ranged from 8 to 23 s.

Materials, procedure and analysis

Lists of two-syllable words were constructed as in the abovementioned experiments. List lengths ranged from three to seven words; trials were discontinued when none of the three sets of a given length was successfully reproduced.

Children were tested individually in a separate room of their school. Memory span was analysed as in the previous experiments.

Results and discussion

There were no systematic effects due to the sequence of the conditions. The word-span means of the three experimental conditions were as follows (standard errors given in parentheses): oral 3.91 (.11); written 3.47 (.12); pointing 3.15 (.14). A repeated measurement ANOVA shows a significant effect of the experimental variation, $F(2, 32) = 16.62$, $MSE = .15$, $p < .001$. *Post-hoc* comparisons show that the observed differences are significant on a .05 level for each pair of conditions. Thus, not only does experiment D provide a replication of the superiority of oral compared to written memory span performance, but the pointing task appeared to be most difficult, decreasing span performance even under the level of writing.

Obviously, our attempt to separate the influences of graphomotoric activity and spelling failed. Rather, the pointing task seems to introduce a kind of secondary task, which impairs memory functions beyond the difficulties involved in handwriting. Although the sequence of the alphabet letters is certainly well learned in fourth grade (and its mastery was demonstrated in the experiment), participants were not used to the alphabet display. Rather, they were presented with a kind of an unpractised keyboard typing condition, where the cognitively costly aspect is not the motor pattern as such, but the target of the finger movement. From this result, a consequence can be derived for all imaginable writing modalities or technologies: Given that the motor-spatial patterns needed to

produce the intended graphical traces are not well practised or at best automated, even simple cognitive tasks will suffer. Particularly keyboard typing needs not only to be taught, but also to be practised (not necessarily in the form of a touch-typing method) in order to prepare students for the subsequent work on writing tasks at the computer.

Unfortunately, we have not been able to contribute decisively to the question of how to successfully separate graphomotoric influences on memory performance from those related to spelling. We have already pointed to the problems of dictation tasks (see also the critical review of Graham et al., 1997, p. 171). Our idea of a motorically simple spelling board apparently introduced an undesirable spatial orientation task; the same probably holds for the use of keyboards of any kind. Of course, the use of any graphical input device could be intensively practised before it is employed in experiments. But that would always also imply spelling training, and thus alter the low-level writing skills to be assessed. Given that early writing education proceeds by handwriting (and that we remain in the worlds of alphabetic or syllable-based scripts), we will have the fact that low-level processes of writing comprise the graphical realization of letters and the letter-by-letter composition of words. From the experiments in adults with unpractised writing modes, we can, after all, conclude indirectly that at least the motor execution component of writing can have an effect on working memory performance.

GENERAL DISCUSSION

The results of the reported experiments clearly support the assumption put forward by Bourdin and Fayol (1994) that for primary school children, written language production is cognitively more costly than oral language production and that the involved difficulty already appears on low-level processes that include orthographic and graphomotoric behaviour. Although this hypothesis relates to cognitive processes that should generally apply to all children of the human species, it is nevertheless necessary to carry out cross-linguistic replications when linguistic materials play an important role in the experimental task. The fact that exactly the same effects as in French emerged for German children, German adults, and German word lists can be considered a strong indication in favour of the given explanations. Particularly, as far as orthography is involved, difficulties relate to the fact that every written word must be spelled somehow, rather than to particular inconsistencies

between phonological and orthographical word forms in French or, more generally, in languages with a rather irregular orthography (Perfetti et al., 1997). Note that the presented experiments do not (and were not intended to) contribute to the question of how best to achieve proper spelling skills. For example, Cunningham and Stanovich (1990) showed that handwritten practice is better for the acquisition of correct spelling than keyboard practice. On the other hand, spelling skills appear to have almost no predictive value for composition fluency and composition quality in primary and intermediate children (Graham et al., 1997). Thus, it is an issue in its own right that the composition of words from letters and the graphic execution of words can have effects on short-term memory performance.

With respect to educational implications, it is particularly interesting that until fourth grade in elementary school, children perform worse on the same task when it is given in the written mode. Since the reported experiments employed only very simple tasks, it is likely that for more complex tasks that involve planning, recall from long-term memory, and idea generation (Torrance & Galbraith, 1999), the written mode will impose even stronger restrictions on children's cognitive resources. Thus, it could be advisable not to rely too much on written performance when the maximum abilities of children are to be assessed. Rather, comparing oral and written performance on the same task could serve as an indication of the progress that a student made in his or her writing skills; the more the written performance on a task equals the oral performance on the same task, the more automated are this student's low-level processes of writing, which is, in turn, an indication of successful writing instruction in primary school.

In general, research on effects of different verbal modes of language production calls attention to the necessity of becoming more sensitive to language-related processes in diagnostic assessment (e.g., Grabowski, Vorweg, & Rummer, 1994, for eyewitness reports). Measures such as working memory capacity or word span can hardly be considered a unique characteristic and a stable feature of a person when they change depending on the verbal mode of assessment. In cases when only differential diagnoses are needed, however, it might be helpful enough to keep the verbal mode at least constant. We observed highly significant ($p < .01$) correlations between the oral and the written versions of memory span measures in the experiments that followed a within-subjects design: $r = .58$ (third-graders, Experiment A),

$r = .41$ (adults, Experiment A), $r = .68$ (fourth-graders, Experiment B), $r = .54$ (fourth-graders, Experiment D). For these age groups, it appears appropriate to conclude that it is only the absolute magnitude of span measures that varies with the verbal output modality, while the relative differences among individuals remain (to the degrees expressed in the correlations) stable across verbal modes. However, for the youngest group in our experiments, the second-graders of Experiment B, no statistically significant correlation between the oral and the written performance occurred ($r = .28$; *ns*). In the early years of literalization, it could be the case that oral and written spans indeed measure different capacities.

There are also questions that remain unanswered in the abovementioned experiments but wait for further clarification. Thus, word span is a task that involves recall from working memory. Here, children do better in the oral than in the written mode, while adults show no difference between the verbal modes. With recall of knowledge from long-term memory, however, a writing superiority effect has frequently been observed for adults (Grabowski, 2007): A given knowledge basis in an individual is more completely diagnosed in the written than in the oral mode of recall. In contrast to the writing superiority effect, which has been mainly demonstrated for list-like knowledge, Kellogg (2007) reports on advantages of the oral recall of a story compared to written accounts. We need to further examine output modality effects in long-term memory tasks, and it looks as if there will be different effects to those we know from working memory tasks.

Moreover, writing subdivides, in a technical world, into handwriting, typing, or other input devices of the present (e.g., pocket computers where you touch display areas which a pen) or the future. What about effects between different ways of writing within the written mode as such? There is preliminary evidence that having adults use either well-practised or unfamiliar writing techniques may influence tasks that depend on working memory capacity (as in experiment C2 above), but does not affect recall from long-term memory (Grabowski, 2007). Similarly, we will still have to find out whether it would facilitate the development of certain writing skills in primary-school children if they began to write by typing on a keyboard—which implies a separation of the complex motor activity of handwriting from spelling skills. We already know, however, that spelling benefits from handwriting rather than typing (Cunningham & Stanovich, 1990). To date, extensive practice of all involved components is

certainly still among the most reliable educational measures to foster linguistic output production as well as memory performance.

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